Convex elicitation of continuous properties

Jessica Finocchiaro, Rafael Frongillo

University of Colorado- Boulder

June 22, 2018

Principal-agent problem

- You want to learn something from a reporter.
- Punish them based on their error.

$$L(r, y)$$

  - r is the report.
  - y is the observed outcome.
- Agent is seeking to minimize their expected loss.

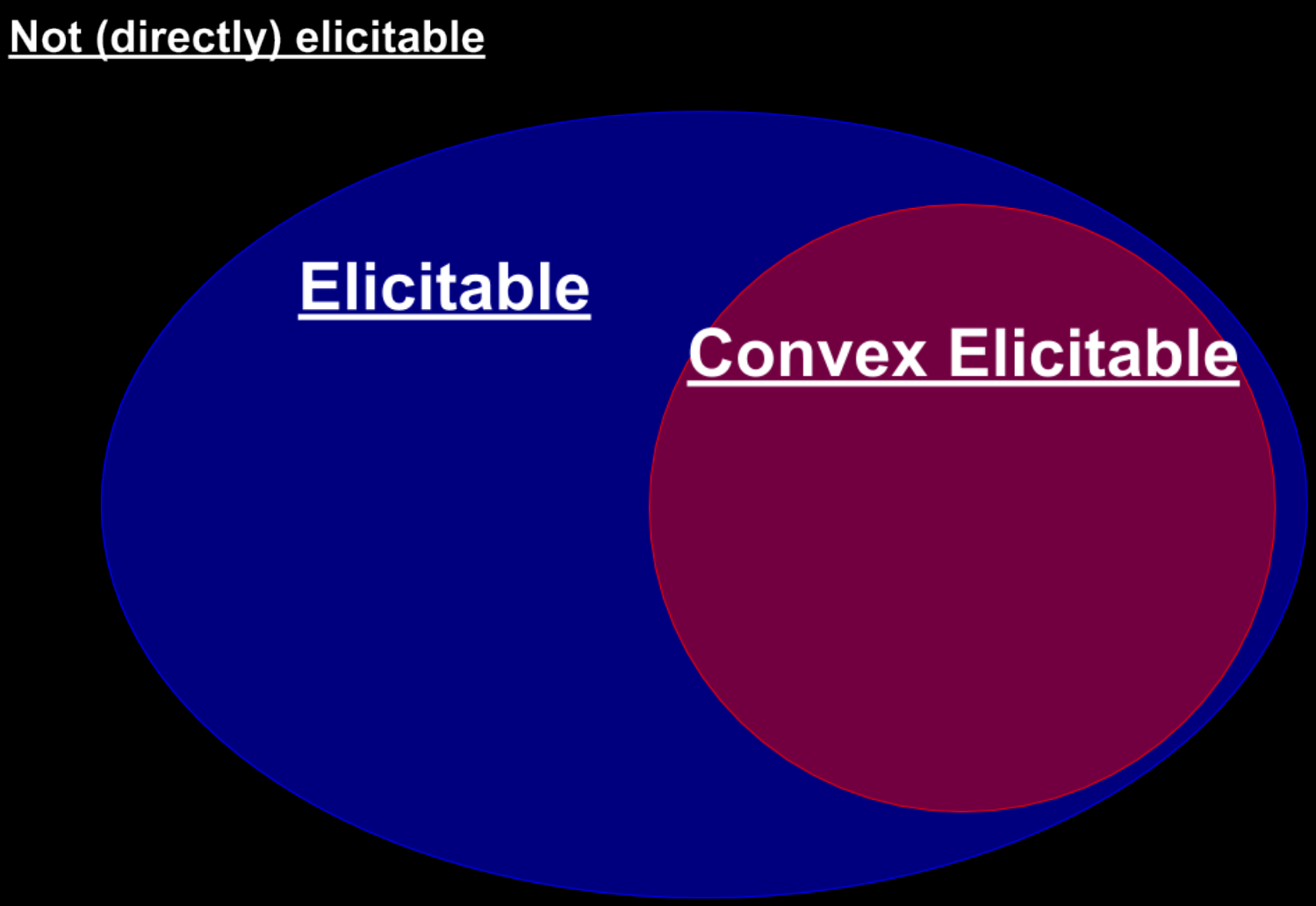
Definitions

- A *property* is a function of a probability distribution $\Gamma : \Delta(\mathcal{Y}) \to \mathbb{R}$
- A loss $L : \mathcal{R} \times \mathcal{Y} \to \mathbb{R}$ *elicits* a property Γ if for all p,

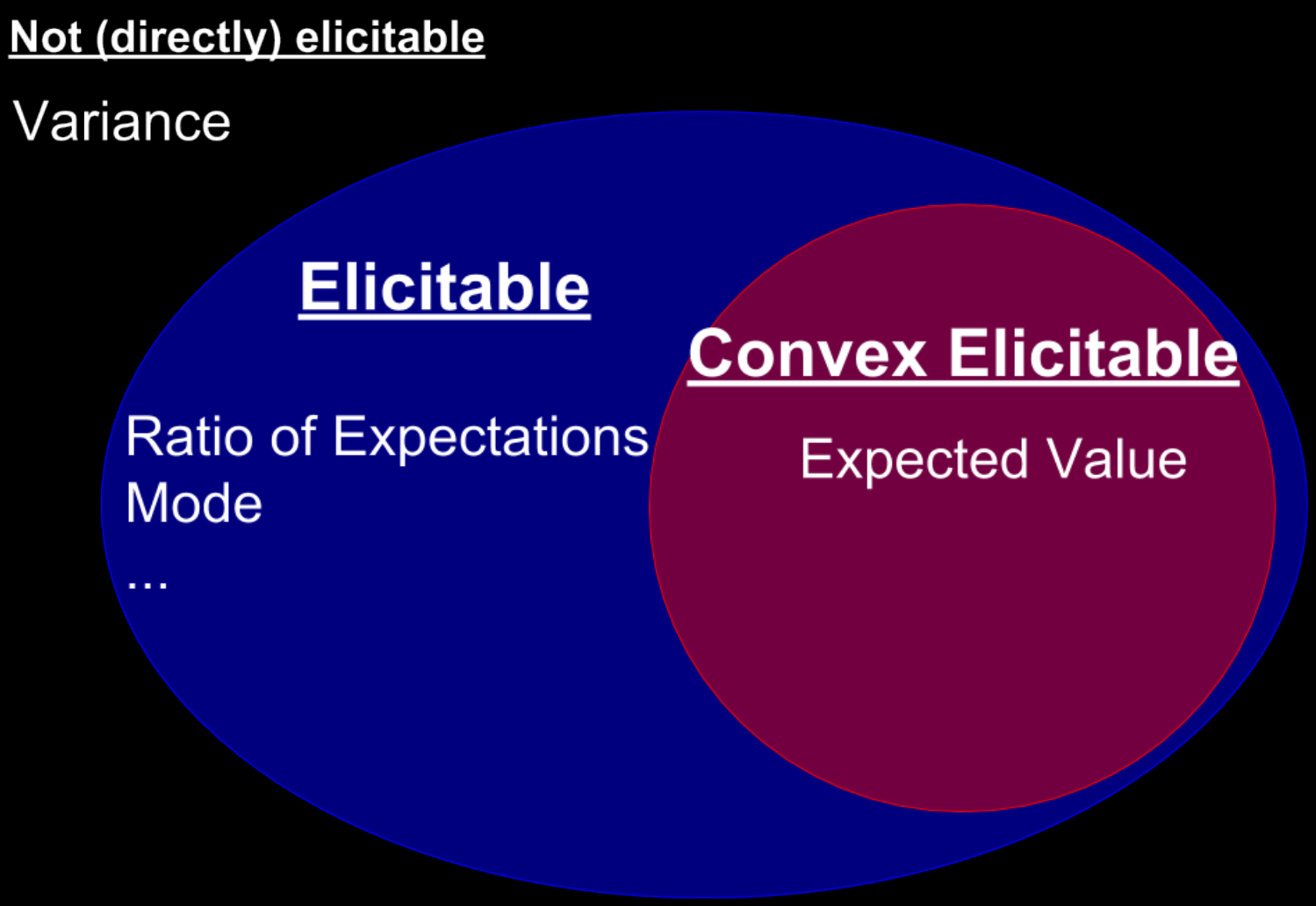
$$\{\Gamma(p)\} = \arg\min_{r \in \mathcal{R}} \mathbb{E}_{Y \sim p} L(r, Y)$$

- Here, $\mathcal{Y}$ is finite

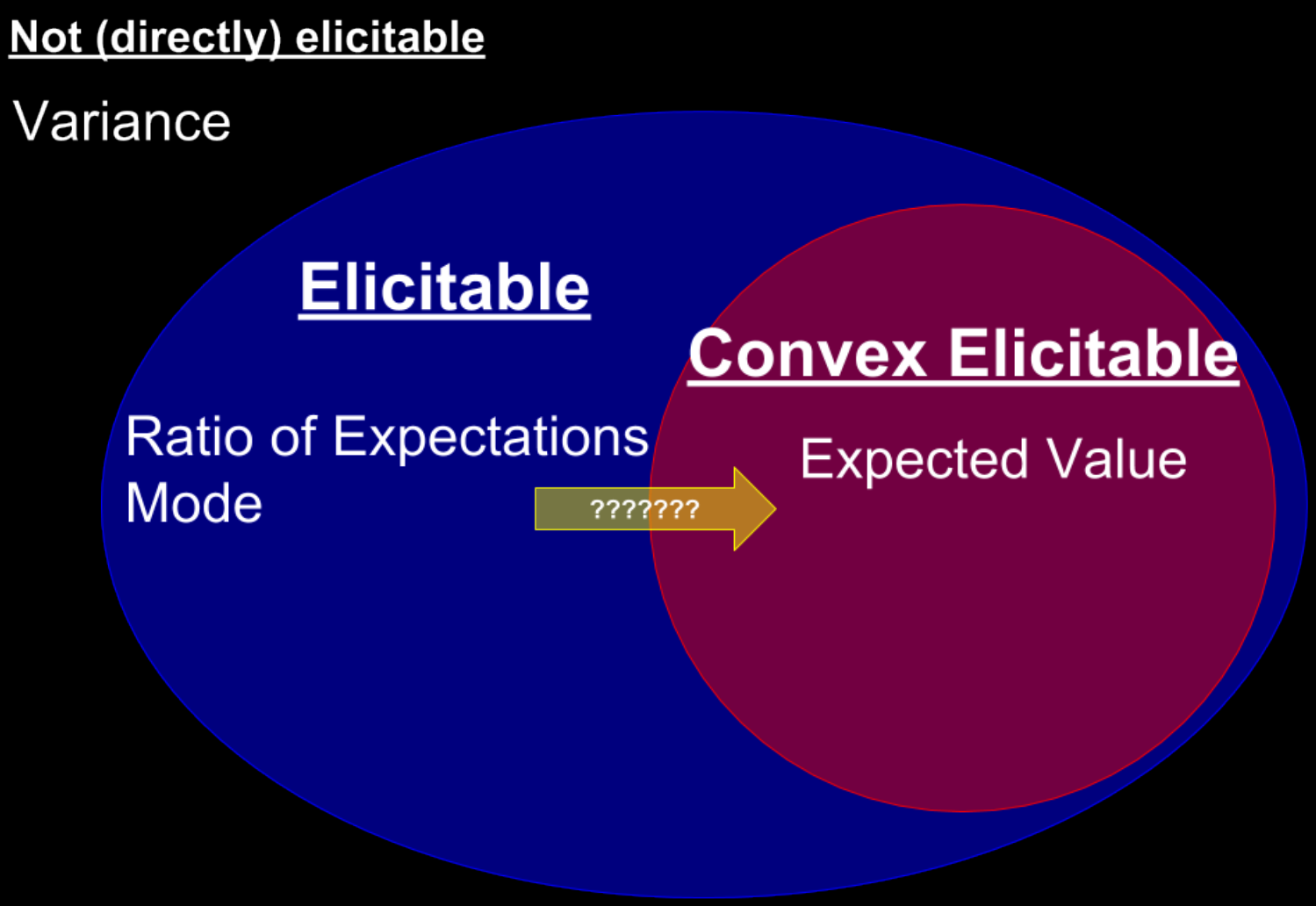
Problem at hand



Problem at hand



Problem at hand



Main result

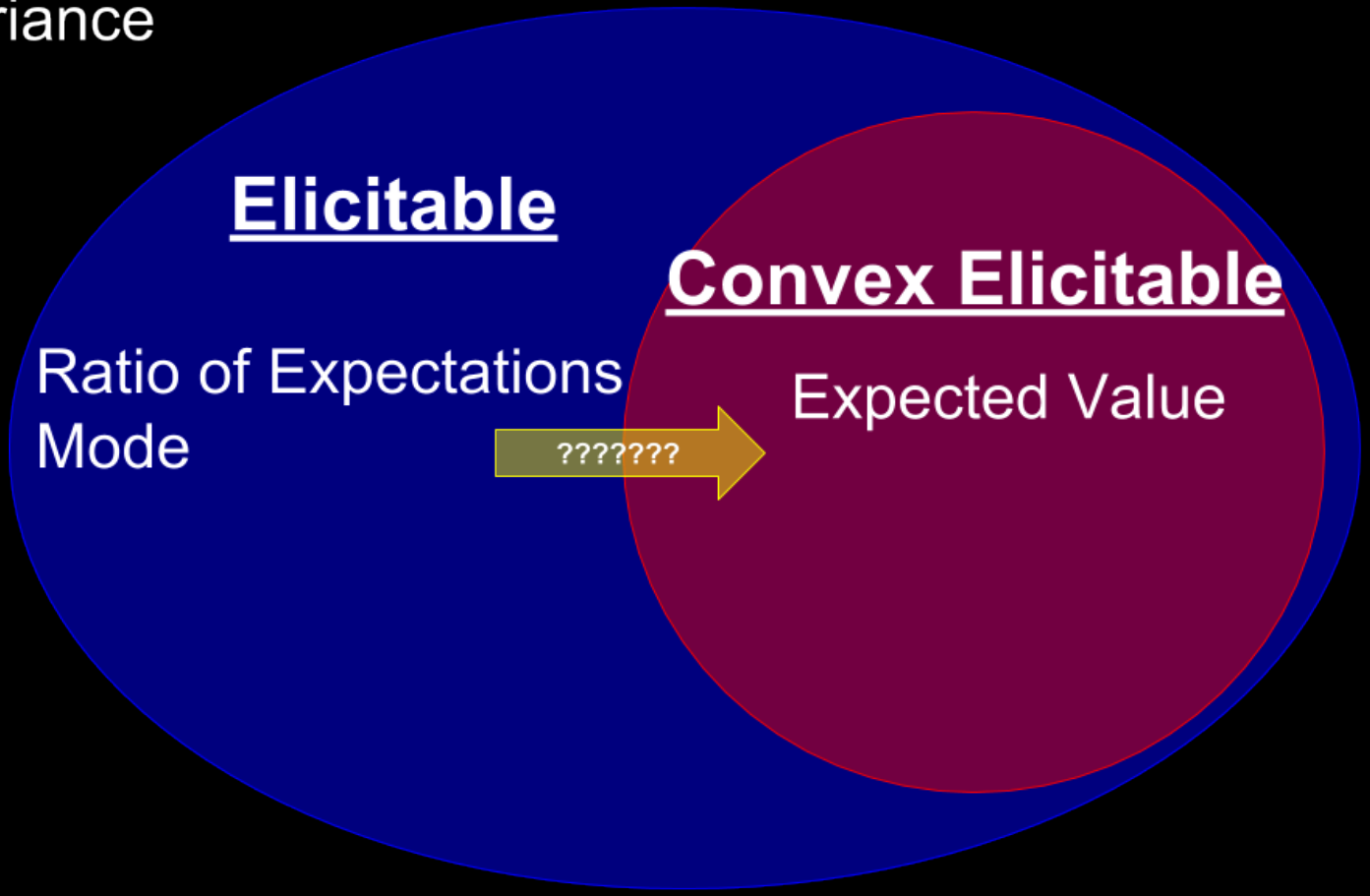
Theorem (Informal)

Let Γ be a real-valued, continuous property.
Then Γ is elicitable $\iff$ Γ is convex elicitable.

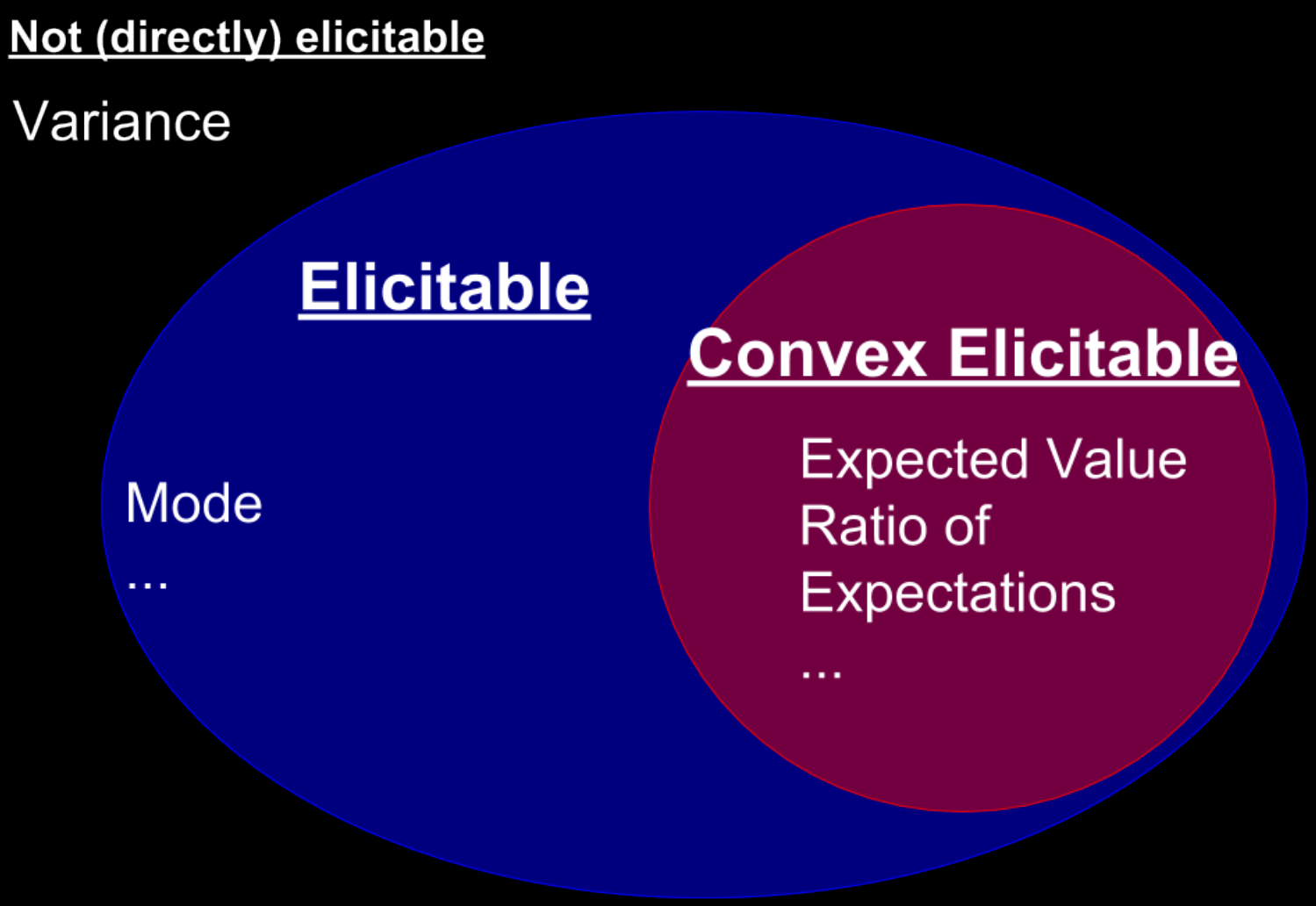
Summary

Not (directly) elicitable

Variance



Summary


Not (directly) elicitable
Variance
Elicitable
Mode
...
Convex Elicitable
Expected Value
Ratio of
Expectations
...

What does this mean to EC?

- Scoring Rule Markets
- Axioms
  - Tractable Trading
  - Bounded Trader Budgets

Thank you

Questions?